Civil Engineering

Calculus and Differential Methods

Professional Learner Notes

Differentiation, integration, ordinary differential equations and numerical methods for civil engineering problems.

Section	Purpose
Rates of change	Gradient and derivative as measures of engineering change.
Differentiation rules	Power, product, quotient and chain rules.
Applications of differentiation	Stationary values, tangents, maxima and minima.
Integration	Area under curves and accumulated quantities.
Differential equations and numerical methods	First-order models, Newton-Raphson and approximate integration.

How to Study This Topic

The learning method follows a programmed engineering mathematics approach: understand the concept, study the formula, follow a worked method, then practise with a technical problem. The aim is not memorisation only; the learner must interpret the answer in the department context.

- Write the formula before substitution.
- Use correct units.
- Check whether the result is reasonable.
- State a technical interpretation.
- Repeat practice until the method is fluent.

Formula Summary

Power rule

$$\frac{d}{dx}(ax^n)=anx^{n-1}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Basic integration

$$\int ax^n\,dx=\frac{ax^{n+1}}{n+1}+C$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

First-order linear ODE

$$\frac{dy}{dx}+P(x)y=Q(x)$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Newton-Raphson

$$x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Trapezoidal rule

$$\int_a^b y\,dx\approx\frac{h}{2}\left[y_0+y_n+2\sum y_i\right]$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

1. Rates of change

Gradient and derivative as measures of engineering change.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

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\frac{d}{dx}(ax^n)=anx^{n-1}
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Worked method

Example: Road profile gradient. For $y = 0.02x^2 + 0.5x + 100$, find the gradient at $x = 20$ m.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports road profiles. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving settlement rate. Show formula, substitution, units and interpretation.

2. Create and solve one technical calculation involving area estimation. Show formula, substitution, units and interpretation.

3. Create and solve one technical calculation involving numerical solving. Show formula, substitution, units and interpretation.

4. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.

5. Create and solve one technical calculation involving flow accumulation. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving settlement rate. Show formula, substitution, units and interpretation.

7. Create and solve one technical calculation involving area estimation. Show formula, substitution, units and interpretation.

8. Create and solve one technical calculation involving numerical solving. Show formula, substitution, units and interpretation.

2. Differentiation rules

Power, product, quotient and chain rules.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\int ax^n\,dx=\frac{ax^{n+1}}{n+1}+C$$

Worked method

Example: Drainage area. Find the area under $y = 0.5x^2 + 1$ from $x = 0$ to $x = 4$.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports flow accumulation. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving area estimation. Show formula, substitution, units and interpretation.

2. Create and solve one technical calculation involving numerical solving. Show formula, substitution, units and interpretation.

3. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.

4. Create and solve one technical calculation involving flow accumulation. Show formula, substitution, units and interpretation.

5. Create and solve one technical calculation involving settlement rate. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving area estimation. Show formula, substitution, units and interpretation.

7. Create and solve one technical calculation involving numerical solving. Show formula, substitution, units and interpretation.

8. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.

3. Applications of differentiation

Stationary values, tangents, maxima and minima.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\frac{dy}{dx}+P(x)y=Q(x)$$

Worked method

Example: Numerical root. Use one Newton-Raphson iteration to approximate a root of $f(x)=x^3-4x-9$ from $x_0=2.5$.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports settlement rate. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving numerical solving. Show formula, substitution, units and interpretation.

2. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.

3. Create and solve one technical calculation involving flow accumulation. Show formula, substitution, units and interpretation.

4. Create and solve one technical calculation involving settlement rate. Show formula, substitution, units and interpretation.

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8. Create and solve one technical calculation involving flow accumulation. Show formula, substitution, units and interpretation.

4. Integration

Area under curves and accumulated quantities.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
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- Interpret the result as a technician, not only as a number.

Main formula for this section

$$x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)}$$

Worked method

Example: Road profile gradient. For $y = 0.02x^2 + 0.5x + 100$, find the gradient at $x = 20$ m.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports area estimation. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

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- Mixing units such as mm, cm and m.
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- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving road profiles. Show formula, substitution, units and interpretation.

2. Create and solve one technical calculation involving flow accumulation. Show formula, substitution, units and interpretation.

3. Create and solve one technical calculation involving settlement rate. Show formula, substitution, units and interpretation.

4. Create and solve one technical calculation involving area estimation. Show formula, substitution, units and interpretation.

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7. Create and solve one technical calculation involving flow accumulation. Show formula, substitution, units and interpretation.

8. Create and solve one technical calculation involving settlement rate. Show formula, substitution, units and interpretation.

5. Differential equations and numerical methods

First-order models, Newton-Raphson and approximate integration.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
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Main formula for this section

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Worked method

Example: Drainage area. Find the area under $y = 0.5x^2 + 1$ from $x = 0$ to $x = 4$.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports numerical solving. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

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Practice set

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8. Create and solve one technical calculation involving area estimation. Show formula, substitution, units and interpretation.

Mixed Revision Questions

1. A civil engineering problem involves flow accumulation. Choose a suitable mathematical method, solve the problem, and explain the result.

2. A civil engineering problem involves settlement rate. Choose a suitable mathematical method, solve the problem, and explain the result.

3. A civil engineering problem involves area estimation. Choose a suitable mathematical method, solve the problem, and explain the result.

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26. A civil engineering problem involves flow accumulation. Choose a suitable mathematical method, solve the problem, and explain the result.

27. A civil engineering problem involves settlement rate. Choose a suitable mathematical method, solve the problem, and explain the result.

28. A civil engineering problem involves area estimation. Choose a suitable mathematical method, solve the problem, and explain the result.

29. A civil engineering problem involves numerical solving. Choose a suitable mathematical method, solve the problem, and explain the result.

30. A civil engineering problem involves road profiles. Choose a suitable mathematical method, solve the problem, and explain the result.

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.

2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.

3. Uploaded KNEC solved mathematics materials for examination-style orientation.

4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.